

Representations, Description and Recognition

Image Processing | CSIT 5th Sem

Yuba Raj Devkota | NCCS

Unit 7	Representations, Description and Recognition	Teaching Hours (5)
Representation and Descriptions	Introduction to some descriptors: Chain codes, Signatures, Shape Numbers, Fourier Descriptors	2 hrs.

Recognition	Patterns and pattern classes, Decision-Theoretic Methods, Introduction to Neural Networks and Neural Network based Image Recognition	2 hrs.
Pattern Recognition	Overview of Pattern Recognition with block diagram	1 hr

Image Representation & Description

- After an image is segmented into regions, the regions are represented and described in a form suitable for computer processing (descriptors).
- Representing a region:
 1. In terms of its external characteristics (boundary)
 2. In term of its internal characteristics

Exp: A region might be represented by the length of its boundary.

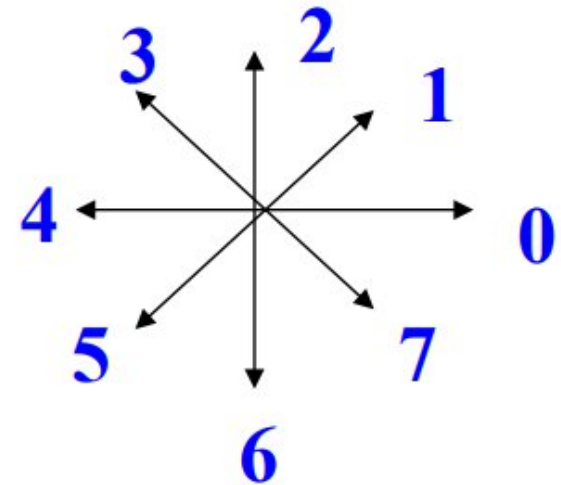
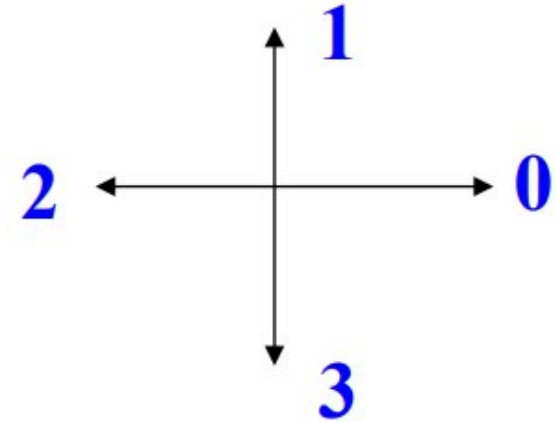
 - External representations are used when the focus is on shape of the region.
 - Internal representations are used when the focus is on color and texture.
 - Representations should be insensitive to rotation and translation.

Chain Code

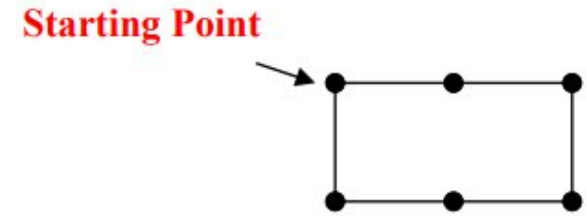
- Chain Code: Used to represent a boundary by a connected sequence of straight line segments.
 - 4 or 8 connectivity is used
 - The direction of each segment is coded by a numbering scheme.

Method:

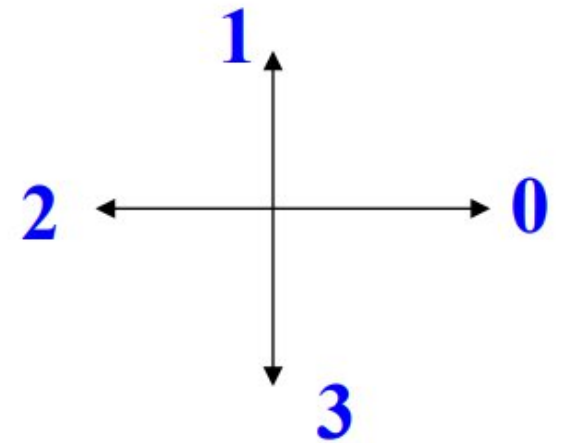
- Follow the boundary in a specific (clockwise) direction.
- Assign a direction to the segment connecting every pair of pixels.



Shape Number

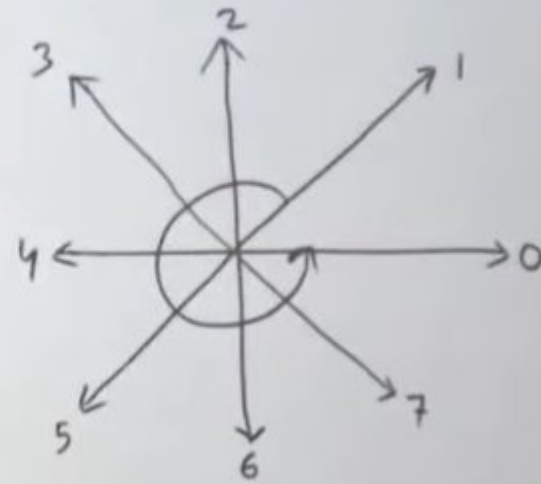
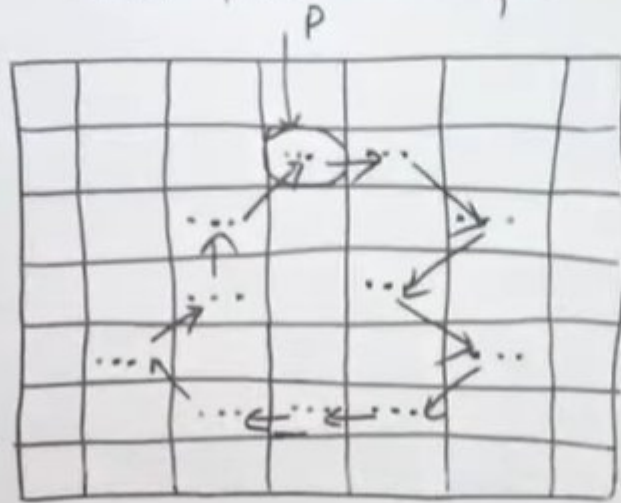


- Shape number: first difference of smallest magnitude (in the chain code)
 - Exp: chain code 003221
 - First difference code: 303303
 - Shape number: 033033



AKTU
20/2-13

Q. Given an image, write down the 8-chain Code and find shape number of it.



8-chain Code

1) Chain Code - 07575443121 ^{8-chain Co}

First-difference - 77626707617

2) Shape no. - 07617776267 ✓

1. Chain Codes

Definition :

A boundary representation technique that encodes the contour of an object as a sequence of directional movements (e.g., 4 or 8 directions). The Freeman Chain Code is the most common, using 8-directional codes (0–7).

Key Properties :

- **Sensitivity** : Prone to noise, rotation, and scaling.
- **Invariance** : Achieved by normalizing the starting point and rotation (e.g., aligning to a dominant direction).

Applications :

- Shape representation, edge detection, and simple boundary analysis.

Pros/Cons :

- *Pros* : Simple, compact, and easy to compute.
- *Cons* : Sensitive to noise and transformations; requires preprocessing for invariance.

Applications Across Domains :

- **Chain Codes** : Align starting point and rotation.
- **Signatures** : Normalize scale and align reference angle.
- **Fourier Descriptors** : Remove DC component, normalize magnitude, and align phase.
- Medical imaging (e.g., cell boundary analysis).
- Robotics (object recognition).
- OCR (character shape matching).

Signature

A **Signature** is a **1-D representation of a 2-D shape boundary**.

Instead of analyzing the entire boundary points of a shape, we convert the shape into a **simpler function or graph** that represents its structure.

Usually, a signature represents the **distance from the centroid (center) of the object to its boundary** at different angles.

So, a complex 2-D shape becomes a **simple curve (function)**.

Steps to Compute Signature

1. Find the **centroid (center point)** of the object.
2. Measure the **distance from centroid to boundary** for different angles (0° – 360°).
3. Plot **distance vs angle**.

This produces a **signature curve**.

Mathematical Representation

If

- $C(x_c, y_c)$ = centroid
- $P(x, y)$ = boundary point

Distance:

$$r(\theta) = \sqrt{(x - x_c)^2 + (y - y_c)^2}$$

where θ is the angle.

Advantages

- Converts **2D shape** → **1D signal**
- Easy for **shape comparison**
- Useful for **object recognition**

Applications

- Character recognition
- Shape matching
- Medical image analysis
- Object detection

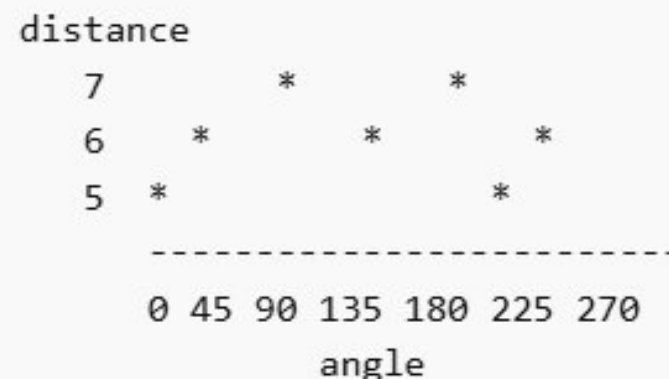
Example

Suppose an object has the following boundary distances from the centroid.

Signature representation:

Angle →	0	45	90	135	180	225	270	315
Distance →	5	6	7	6	5	6	7	6

Graph:



This **curve** represents the shape.

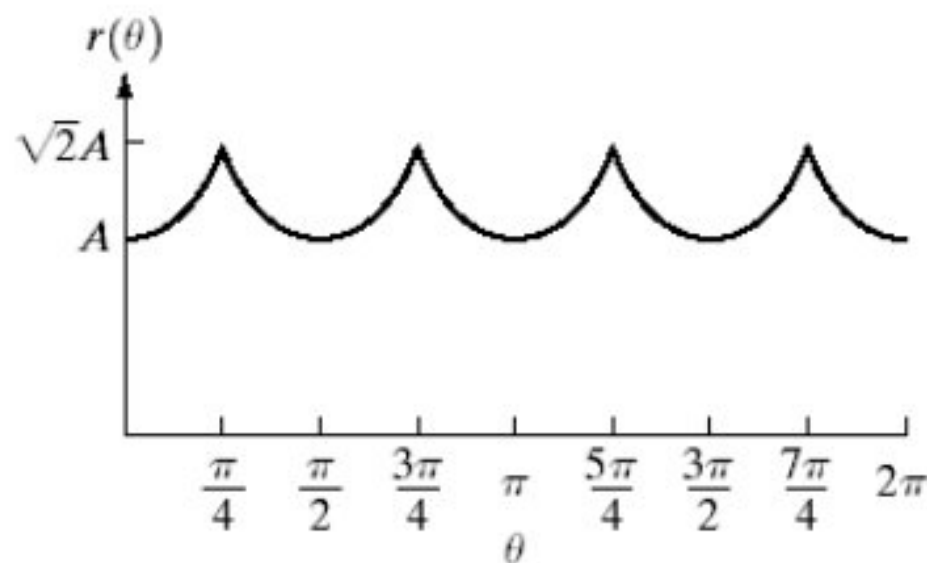
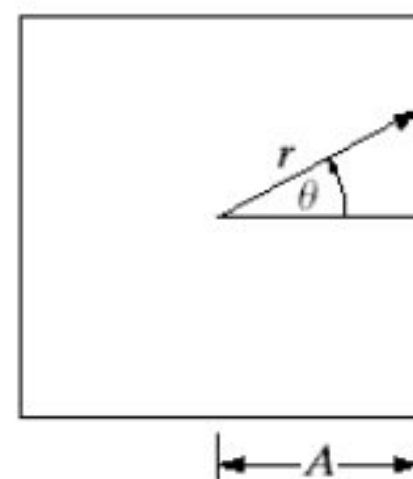
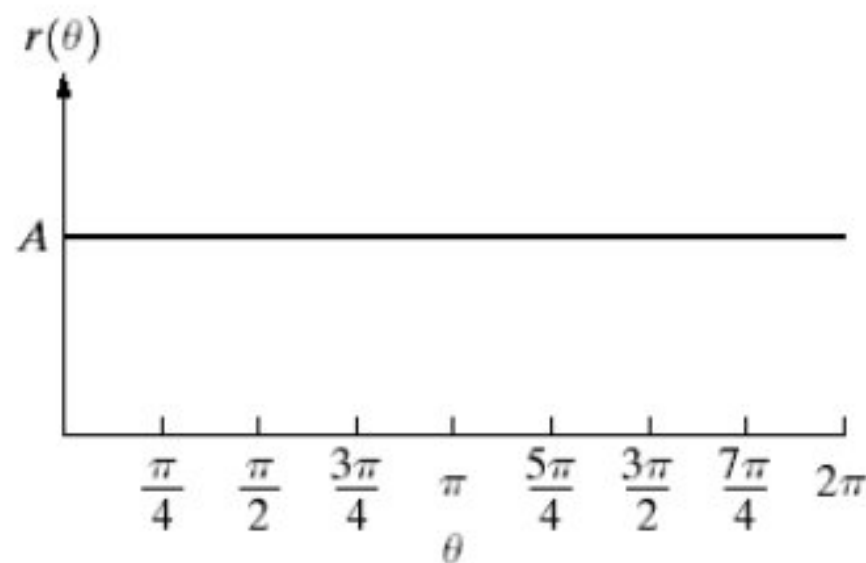
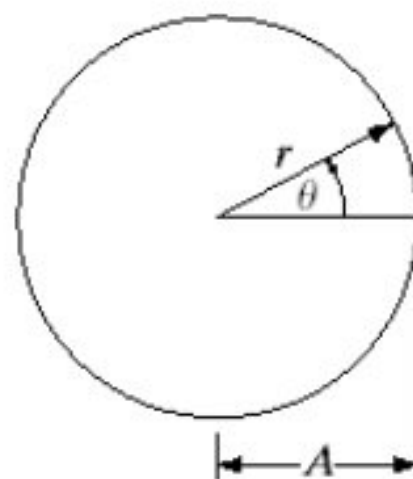
a b

FIGURE 11.5

Distance-versus-angle signatures.

In (a) $r(\theta)$ is constant. In (b), the signature consists of repetitions of the pattern

$r(\theta) = A \sec \theta$ for $0 \leq \theta \leq \pi/4$ and $r(\theta) = A \csc \theta$ for $\pi/4 < \theta \leq \pi/2$.



Fourier Descriptor

Fourier Descriptors (FD) describe the **shape** of an object using **Fourier Transform**.

The boundary of an object is treated as a **sequence of points**, and **Fourier Transform** converts it into **frequency components**.

Fourier Transform

These frequency components are called **Fourier Descriptors**.

They represent **shape information in compact form**.

$$F(u) = \frac{1}{N} \sum_{k=0}^{N-1} z(k) e^{-j2\pi uk/N}$$

The values $F(u)$ are called **Fourier Descriptors**.

Boundary Representation

If the boundary has points:

$$(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots, (x_{N-1}, y_{N-1})$$

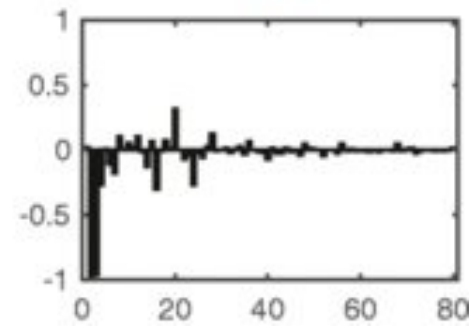
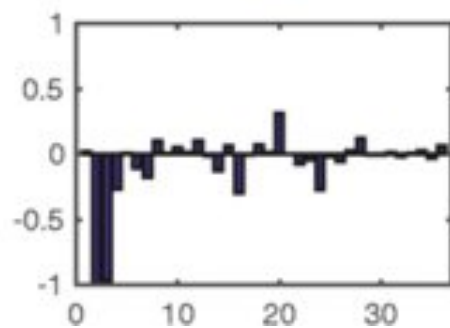
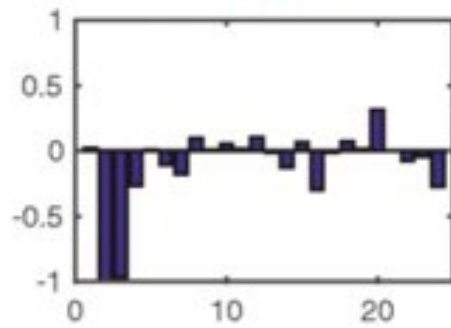
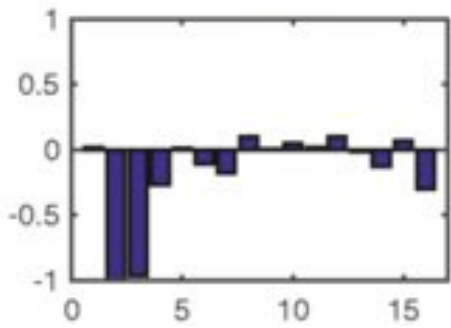
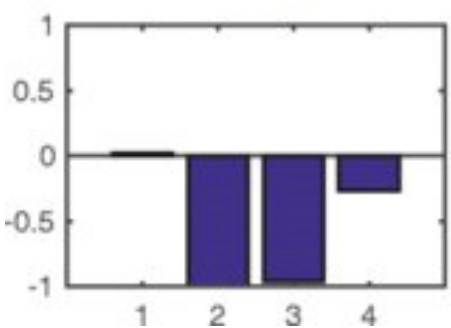
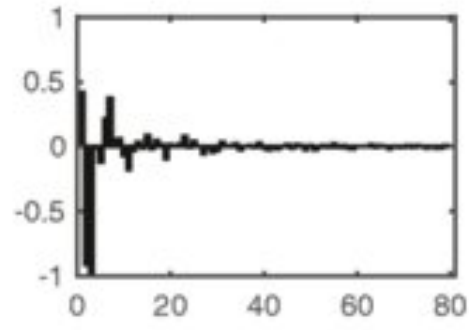
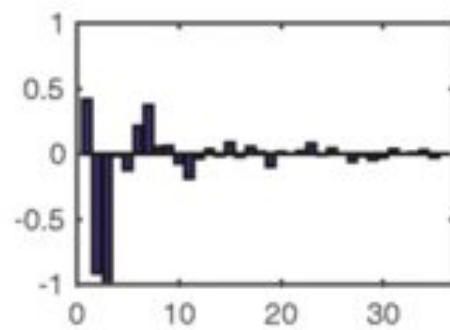
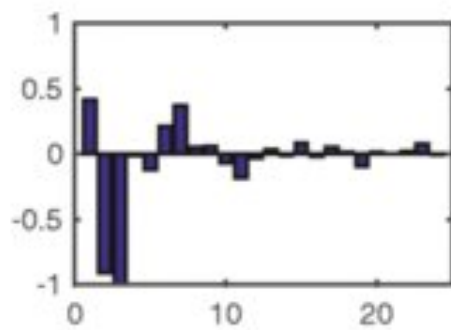
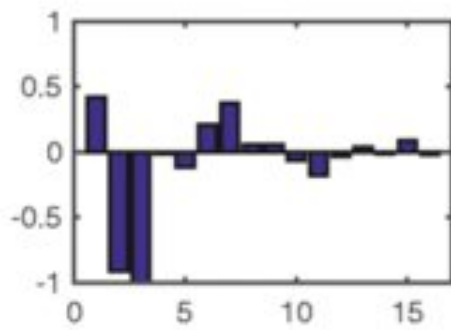
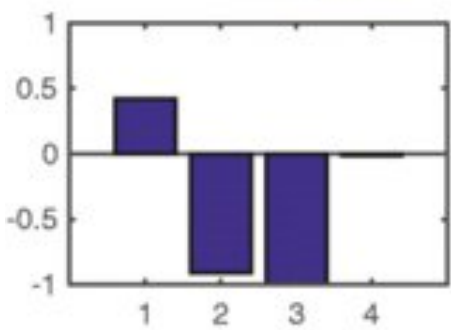
We convert them into **complex numbers**:

$$z(k) = x(k) + jy(k)$$

where

$$j = \sqrt{-1}$$

- *Fourier descriptors represent a shape as frequency components.*
- *Low frequencies describe the general shape, while high frequencies describe fine boundary details.*



Pattern Recognition System

Pattern

A **pattern** is a set of features or characteristics that describe an object or data. It represents the **structure or arrangement of data** that can be identified or classified.

Examples:

- A handwritten digit (0–9)
- Fingerprint image
- Speech signal
- Face image

In simple words, a **pattern** is the representation of an object using measurable features.



Pattern Recognition

Pattern Recognition is the process of **identifying or classifying patterns** in data using **algorithms or statistical techniques**.

It automatically detects regularities in data and assigns them to specific categories.

Examples:

- Face recognition in mobile phones
- Handwritten digit recognition
- Speech recognition (Google Assistant, Siri)
- Medical image diagnosis

Steps of Pattern Recognition System

A typical **Pattern Recognition System** follows these steps:

1. Data Acquisition

Data or images are collected using sensors, cameras, scanners, microphones etc.

Example:

A camera captures an image of a handwritten digit.

2. Pre-processing

The collected data is cleaned and improved to remove noise.

Common operations:

- Noise removal
- Image enhancement
- Normalization
- Filtering

Example:

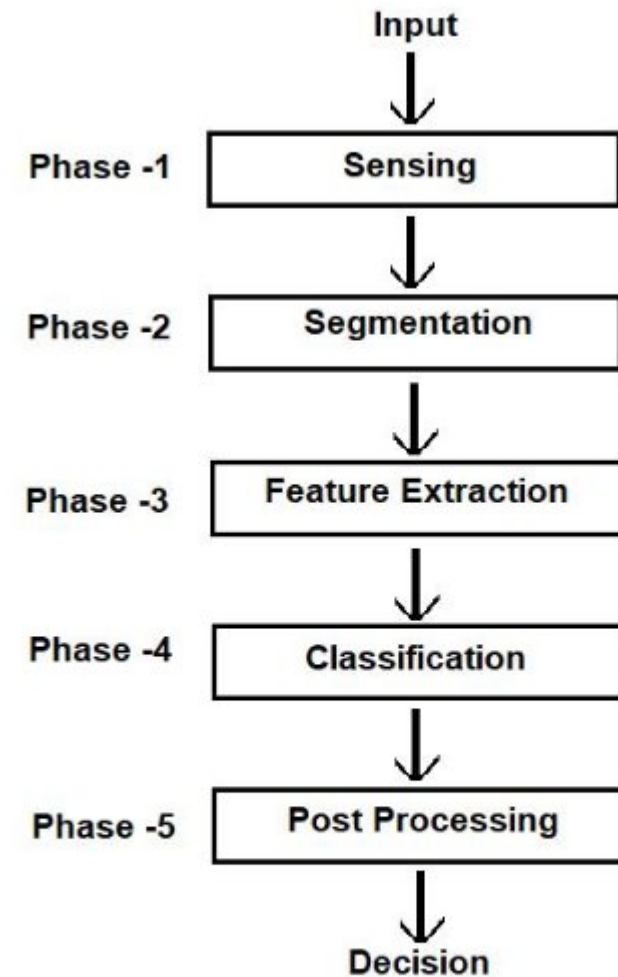
Removing noise from the handwritten digit image.

3. Segmentation

The image is divided into meaningful regions or objects.

Example:

Separating a digit from the background.



Phases in Pattern Recognition

4. Feature Extraction

Important characteristics of the object are extracted.

Examples of features:

- Shape
- Edges
- Texture
- Color
- Corners

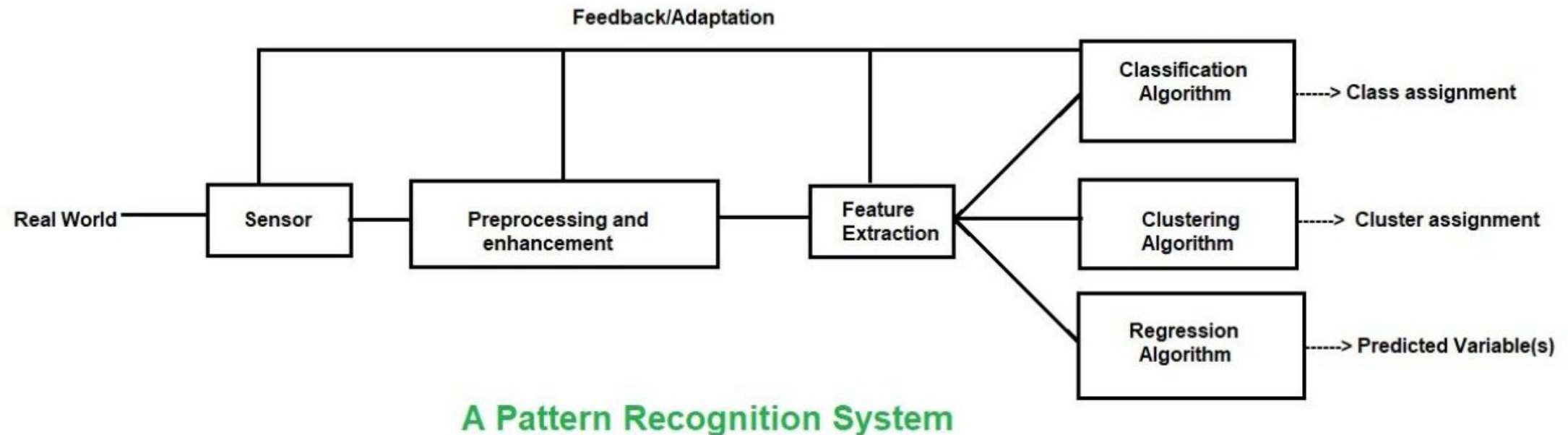
This step reduces data while keeping useful information.

5. Classification / Recognition

The extracted features are compared with known patterns stored in a database and classified.

Example:

The system decides whether the digit is **3, 5, or 8**.



Application Areas of Pattern Recognition

Pattern recognition is used in many fields:

1. Image Processing

- Face detection
- Object detection
- Medical image analysis

2. Speech Recognition

- Voice assistants
- Speech-to-text systems

3. Handwriting Recognition

- Automatic reading of handwritten numbers
- Postal code recognition

4. Biometrics

- Fingerprint recognition
- Face recognition
- Iris recognition

5. Medical Diagnosis

- Tumor detection in MRI/CT scans
- Disease identification

6. Industrial Automation

- Quality inspection
- Defect detection in manufacturing

Example: Detecting Handwritten Digit "0"

Step 1: Input Image

A scanner captures an image of a handwritten number.

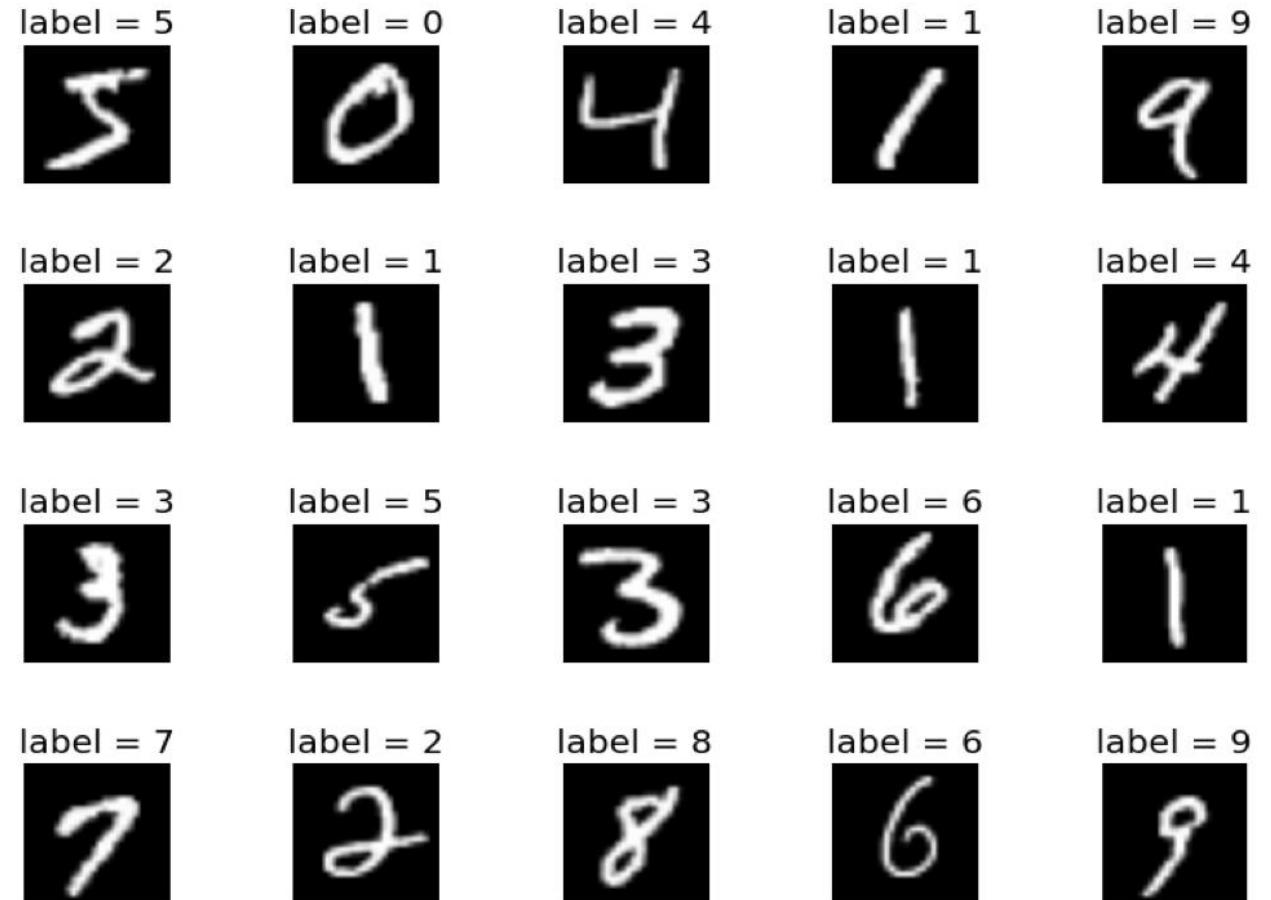
Step 2: Preprocessing

- Convert image to grayscale
- Remove noise
- Convert to binary image

Step 3: Feature Extraction

Features are extracted such as:

- Circular shape
- Closed boundary
- Number of endpoints = 0



Step 4: Classification

The system compares the features with stored patterns.

If the features match the stored model of **digit 0**, the system classifies it as:

Output: Digit = 0

✓ Summary

Pattern recognition is the process of identifying patterns in data.

The system typically follows these steps:

1. Data Acquisition
2. Pre-processing
3. Segmentation
4. Feature Extraction
5. Classification

It is widely used in **AI**, image processing, biometrics, speech recognition, and medical diagnosis.

3. Introduction to Neural Networks

Definition :

Computational models inspired by biological neurons, capable of learning complex patterns via layered architectures.

Structure :

- **Layers** : Input, hidden, and output layers.
- **Neurons** : Compute weighted sums + activation functions (e.g., ReLU, sigmoid).
- **Training** : Adjusts weights using **backpropagation** and **gradient descent** .

Pros/Cons :

Example :

- **Multilayer Perceptron (MLP)** :
 - Input: Image pixels.
 - Hidden layers: Learn hierarchical features.
 - Output: Class probabilities (e.g., "cat" or "dog").
- *Pros* : Automatically learns features; excels at complex tasks (e.g., image recognition).
- *Cons* : Requires large datasets and computational resources.

Key Concepts :

- **Deep Learning** : Networks with many hidden layers (deep networks).
- **Convolutional Neural Networks (CNNs)** : Specialized for grid-like data (e.g., images) via convolutional layers.

4. Neural Network-Based Image Recognition

Definition :

Use of neural networks (especially CNNs) to detect, classify, or segment objects in images.

How It Works :

- **CNN Architecture :**
 - **Convolutional Layers :** Extract local features (edges, textures).
 - **Pooling Layers :** Reduce spatial dimensions (e.g., max pooling).
 - **Fully Connected Layers :** Classify based on learned features.

Examples :

- **Facial Recognition :** Detects faces using CNNs trained on labeled datasets (e.g., VGGFace).
- **Object Detection :** Models like YOLO or Faster R-CNN localize and classify objects.
- **Medical Imaging :** Identifies tumors in X-rays using U-Net (segmentation).

Key Applications :

- Autonomous vehicles (road sign detection), retail (product recognition), agriculture (crop disease detection).

Pros/Cons :

- *Pros :* State-of-the-art accuracy; handles noise and variations.
- *Cons :* Computationally expensive; requires labeled training data.

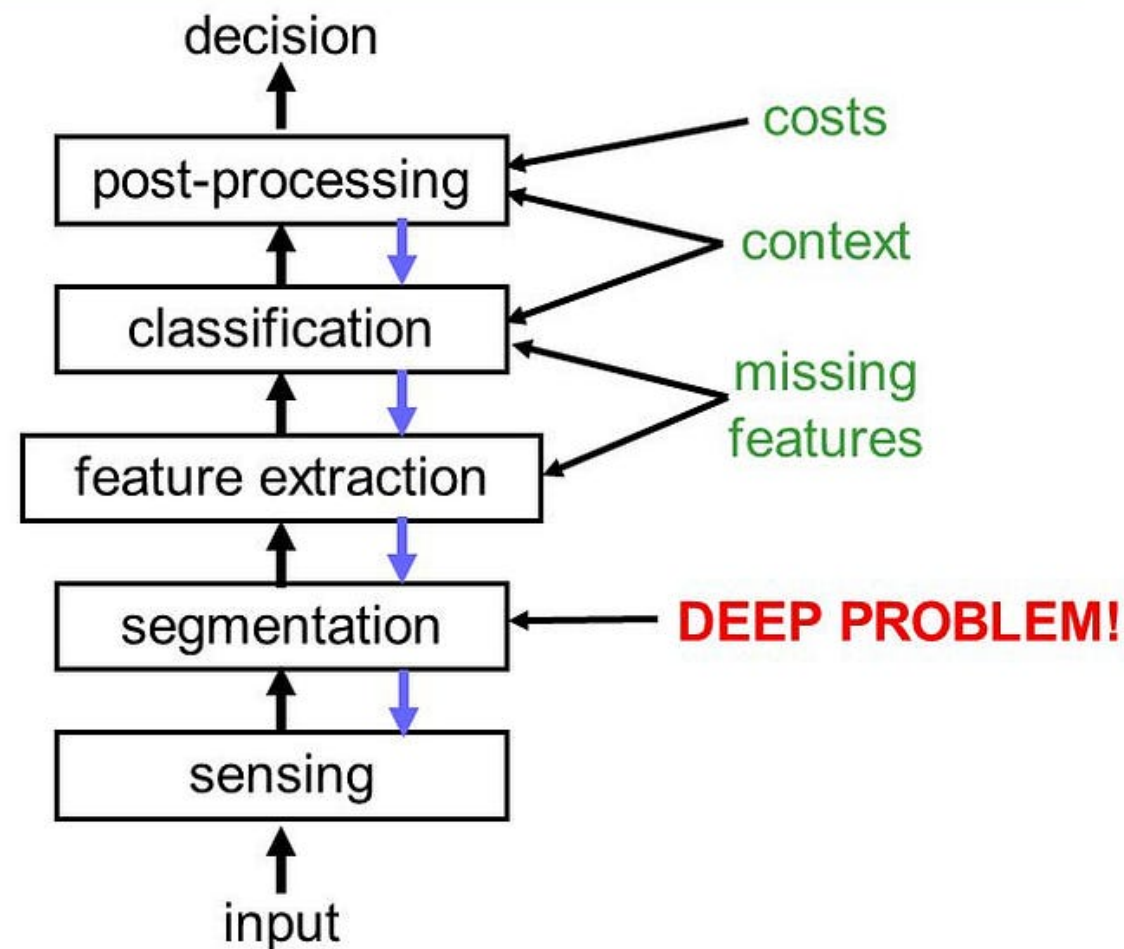
Summary Table

TOPIC	TYPE	KEY EXAMPLE	STRENGTHS	WEAKNESSES	↴
Patterns/Pattern Classes	Feature-based	MNIST digit classification	Interpretable, flexible	Manual feature engineering	
Decision-Theoretic	Statistical	Medical diagnosis using Bayes' theorem	Works with small data	Assumes model correctness	
Neural Networks	Model-based	MLP for image classification	Learns complex patterns	Data-hungry, black-box	
NN Image Recognition	Application	CNNs for facial recognition	High accuracy, robust to noise	Requires GPUs, large datasets	

Block Diagram Explanation:-

1. The sensor converts images or sounds or other physical input into signal data.
2. Segmentation is the part which enables segregating sensed object from the background noise.
3. The feature extractor measures object properties and select those features that are important for classification.
4. The classifier uses these features to assign the sensed object to a category.
5. Finally, the post-processor can take account of other considerations such as the cost of the error to decide appropriate action.

General Structure of a Pattern Recognition System



Applications of Pattern Recognition

- **Image processing, segmentation, and analysis:** Pattern recognition is used to give human recognition intelligence to machines that are required in image processing.
- **Computer vision:** Pattern recognition is used to extract meaningful features from given image/video samples and is used in computer vision for various applications like biological and biomedical imaging.
- **Seismic analysis:** The pattern recognition approach is used for the discovery, imaging, and interpretation of temporal patterns in seismic array recordings. Statistical pattern recognition is implemented and used in different types of seismic analysis models.
- **Radar signal classification/analysis:** Pattern recognition and signal processing methods are used in various applications of radar signal classifications like AP mine detection and identification.
- **Speech recognition:** The greatest success in speech recognition has been obtained using pattern recognition paradigms. It is used in various algorithms of speech recognition which tries to avoid the problems of using a phoneme level of description and treats larger units such as words as pattern
- **Fingerprint identification:** Fingerprint recognition technology is a dominant technology in the biometric market. A number of recognition methods have been used to perform fingerprint matching out of which pattern recognition approaches are widely used.

Explain how can a neural network be applied in digital image processing with the help of a simple perceptron

- Neural networks, particularly simple perceptron's, can be applied in digital image processing to perform tasks like image classification, edge detection, and noise reduction. Here's a detailed explanation of how a simple perceptron can be used in digital image processing:

Understanding the Simple Perceptron

- A simple perceptron is the most basic type of artificial neural network. It consists of a single layer of neurons (also called nodes) that are fully connected to the input layer. Each neuron performs a weighted sum of its inputs, applies an activation function, and produces an output.

Steps to Apply a Simple Perceptron in Digital Image Processing

1. Image Representation:

1. Digital images are represented as matrices of pixel values. For a grayscale image, each pixel value ranges from 0 (black) to 255 (white).
2. For simplicity, consider a 3x3 grayscale image, which can be represented as a 9-element vector.

1. Input Layer:

1. The input layer of the perceptron will have one neuron for each pixel in the image. For the 3x3 image, there will be 9 input neurons.

2. Weights and Bias:

1. Each input neuron is connected to an output neuron via a weight. Additionally, a bias term is added to the weighted sum to shift the activation function.
2. Initialize weights and biases, usually with small random values.

4. Weighted Sum:

- Each input pixel value is multiplied by its corresponding weight, and the results are summed up along with the bias term. Mathematically, this is represented as:

$$z = \sum_{i=1}^n w_i \cdot x_i + b$$

where x_i are the pixel values, w_i are the weights, b is the bias, and z is the weighted sum.

5. Activation Function:

- The weighted sum z is passed through an activation function. Common activation functions include the step function, sigmoid, and ReLU (Rectified Linear Unit).
- For a simple perceptron, a step function can be used to produce a binary output:

$$\text{output} = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$$

6. Output:

- The output of the perceptron can be interpreted as a class label in image classification tasks. For example, a perceptron can be trained to distinguish between images of cats and dogs.

Example: Edge Detection

Let's consider a simple edge detection example using a 3x3 image and a perceptron.

1. Input Image:

```
Copy code

1 1 1
0 0 0
1 1 1
```

Weight Initialization: Initialize weights to detect horizontal edges, for instance

```
CSS

w = [1, 1, 1, -1, -1, -1, 1, 1, 1]
```

3. Bias:

- Set the bias to a small value, e.g., $b = 0$.

4. Weighted Sum Calculation:

- Flatten the 3x3 image to a 9-element vector and compute the weighted sum.

$$z = (1 \cdot 1) + (1 \cdot 1) + (1 \cdot 1) + (0 \cdot -1) + (0 \cdot -1) + (0 \cdot -1) + (1 \cdot 1) + (1 \cdot 1) +$$

$$z = 1 + 1 + 1 + 0 + 0 + 0 + 1 + 1 + 1 = 6$$

5. Activation Function:

- Apply the step function:

$$\text{output} = \begin{cases} 1 & \text{if } 6 \geq 0 \\ 0 & \text{if } 6 < 0 \end{cases}$$

- The output is 1, indicating a detected edge.

Conclusion

- A simple perceptron can be effectively used in digital image processing tasks such as edge detection. By carefully selecting and adjusting the weights and biases, a perceptron can learn to recognize patterns and features in images. For more complex tasks, multi-layer perceptron's (MLPs) or deep neural networks (DNNs) are used, as they can capture more intricate patterns and relationships in the data.

- Explain how can a neural network be applied in digital image processing with the help of a simple perceptron
- Discuss Neural Network based image recognition system with the help of a simple perception.
- What are components of a pattern recognition system? Explain the relevant diagrams.
- Write short notes on:(3 + 3)
 - a) Pattern Recognition and its applications
 - b) Neural Network
- What is pattern and pattern recognition? Explain the steps and application areas of pattern recognition system.
- What is pattern recognition? Explain the applications of neural networks in pattern recognition.